

The Dataverse Network: An Infrastructure for Data Sharing

Gary King
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(8/14/08 talk at “UseR! 2008”, Technische Universität, Dortmund, Germany)

- Gary King, **An Introduction to the Dataverse Network as an Infrastructure for Data Sharing**, *Sociological Methods and Research*, 32, 2 (November, 2007): 173–199.

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- Micah Altman and Gary King. **A Proposed Standard for the Scholarly Citation of Quantitative Data**, *D-Lib Magazine*, 13, 3/4 (March/April, 2007).

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- More information: <http://TheData.org>

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 - uncertain, time consuming, annoying, error prone

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- We propose: **technological solutions to these political problems**

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 - Solution must not require lawyers (we've automated the IRB)

Rules for Citing Printed Matter

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First author (last name first)

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Citations: rule-based, precise, redundant

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Print Citations Work: authors don't think publishers get all the credit; cited articles can be found; copyeditors don't need to see the original to know it exists; the link from citation to print persists

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- Cryptographic technology: **any** change in data content changes the UNF. (cannot tinker after the fact!)
- Noninvertible properties
 - UNFs convey no information about data content
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 - Copyeditor can validate data's existence even without authorization
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- Future researchers can quickly check that they have the same data as used by the author: merely recalculate the UNF

Web 2.0 Terminology

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<http://www.peterson.com>

<http://dvn.iq.harvard.edu/peterson>

Bob Peterson
Professor of Government

Blog: Why the OSA is wrong this time on open-source projects
January 30, 2008, 9:37 AM PST

"More open-source projects have originated in Europe than anywhere else in the world," said Bertrand Dant, chief executive officer of data-integration specialist Talent, a founding member of the Open Solutions Alliance. Founded a year ago, the OSA has had a U.S. focus until now.

The European chapter of the OSA will be formally incorporated in the next 90 days, and will then look for interoperability work required by European users.

"When the OSA got started, we saw our mission as a global one, but the critical mass of activity so far has been in the U.S.," said Dominic Sartorio, OSA president and senior director of product management at services company SpikaSource. "We were approached two months ago by a group of companies in Europe thinking of forming a group like ours but focused on Europe. This led us to think that while we are trying to achieve results globally, there are some parts of the world where we are doing less well."

Sartorio sees 2008 as turning point for open source. In late 2007, he promised that OSA would "out-Microsoft Microsoft" in response to a user survey that revealed a demand for interoperability among

Current Project: Political Economy in Ghana

In the last 20 years the economics profession has been revolutionized by political economy. For example, until the mid 1980s in traditional public finance or macroeconomics, the role of economists was to design socially optimal policies faced with various types of trade-offs. The fact that actual observed policies deviated, often radically, from those proposed by economists was not systematically explained other than by the implicit notion that policymakers did not understand what the best policy was. In addition, many explanations for underdevelopment went in terms of market failures, such market failures were taken as exogenous and their implications derived.

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- Replication data for: On Party Platforms, Mandates, and Government Spending by Gary King, Michael Laver

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REPLICATION DATA FOR: IMPROVING FORECASTS OF STATE FAILURE

Cataloging Information Documentation, Data and Analysis

Citation Information

How to Cite

Study Global ID

Authors

Production Date

Distributor

Distributor Contact

Deposit Date

Replication For

Provenance

Abstract and Scope

We offer the first independent scholarly evaluation of the claims, forecasts, and causal inferences of the State Failure Task Force and their efforts to forecast when states will fail. State failure refers to the collapse of the authority of the central government to impose order, as in civil wars, revolutionary wars,

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Cataloging Information Documentation, Data and Analysis

Download all files in a single archive file (files that you cannot access will not be downloaded)

File Name	Description	Case/Variables	Type	Controls
1. Documentation	Articles related to this study: Improving Forecasts of State Failure		application/pdf	
2. Replication Data	Sample spec file for committee n's			
nm0.apic	For use with committee n's, ASCII data file			
nm.dat	For use with committee n's	7190 11	Tab delimited	
nm.tab	Additional document describing the format of the replication files			
replication.txt				
3. Recoded Data	Country names, in original State format			
names.dta	Country Names Data	8580 3	Tab delimited	
names.tab	Recoded Full Data in Original Format, comma delimited text file			
sort1 full.dta	Documentation for Original		application/pdf	



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REPLICATION DATA FOR: CONSTITUENCY SERVICE AND INCUMBENCY ADVANTAGE
DATA FILE: CONSTITUENCYSERVICE.TAB [Back to Study](#)

[Download Subset](#) [Subset and Recode](#) [Descriptive Statistics](#) [Advanced Statistical Analysis](#)

Selected Variables

To recode (subset) a variable into a different one, first, select a variable from the selected-variables box, click the arrow button below, and then the name and label of the variable you have chosen appear in the new variable name and label boxes for convenience. You must replace the old variable name with a unique variable name that is not used in the data file; the new variable label is optional and you can leave it blank.

Select variables from table below (selected variables will be displayed above) You do not have access to the subsetting and analysis functionality for this restricted data file. You can only view the variables and their summary statistics.

Show 20 Variables ▾

Variable Type	Variable Name/Variable Label	Quick Summary
☐ Continuous	INCAD incumbency advantage	View
☐ Continuous	BUDG budget figures	View
☐ Continuous	SAL salary	View
☐ Discrete	CO state level dummy variables	View

http://hsarewebdata.com

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Download Subset Subset and Recode Descriptive Statistics Advanced Statistical Analysis

Selected Variables

Choose a Statistical Model—
Choose a Statistical Model—

Categorical Data Analysis
Cross-Tabulation

Ecological Inference Models
Hierarchical Multinomial–Dirichlet Ecological Inference Model for R x C Tables

Selected Event Count Models
variables are

- Negative Binomial Regression for Event Count Dependent Variables
- Poisson Regression for Event Count Dependent Variables
- Generalized Additive Model for Event Count Dependent Variables
- General Estimating Equation for Poisson Regression
- Social Network Poisson Regression for Event Count Dependent Variables

Variable

- Continuous Models for Continuous Bounded Dependent Variables
 - Cox Proportional Hazard Regression for Duration Dependent Variables
 - Exponential Regression for Duration Dependent Variables
 - Gamma Regression for Continuous, Positive Dependent Variables
 - General Estimating Equation for Gamma Regression
- Discrete
 - CT state level dummy variables
 - Discrete DE state level dummy variables
 - Discrete IA state level dummy variables
 - Discrete MI state level dummy variables
 - Discrete MO state level dummy variables
 - Continuous SAL salary
 - Discrete CO state level dummy variables

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The Dataverse Network Project Homepage (<http://TheData.org>)

Dataverse Networks may harvest metadata from each other (dashed lines)

The IQSS Dataverse Network at Harvard University

SMR Dataverse

Gary King's Dataverse

PoliSci 101 Dataverse

U.S. Census Dataverse

APSA Legislative Section
Dataverse

NSF Dataverse

Weatherhead Center
Dataverse

MacArthur Network on
Inequality Dataverse

Dept of Psychology
Dataverse

[Other] Dataverse Network[s]

Each Dataverse Network may serve numerous individual dataverses (arrows)

Most users come directly to a dataverse, which is a self-contained archive (with all services provided by a Dataverse Network)

Software is available at the project homepage and only needs to be installed to establish a Dataverse Network. Dataverses are virtual hosts.

Dataverse owners may list, or allow searching across, other dataverses or the data sets in them

Your Dataverse

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- **Results**: **Articles with data available have twice the impact factor!**
(with dataverse, it should be more)

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- Future Researchers: browse or search for a dataverse or dataset; forward citation search; verification via UNFs; subsetting; read metadata, abstract, & documentation; check for new versions; translate format; statistical analyses; download

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- (GUI time not wasted: save R code for replication or further analysis)

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<http://TheData.org>

Technology used in DVN Software

- Language: **Java Enterprise Edition 5** (with EJB3 and JSF) (team picked for JavaOne; Sun engineers regularly call for advice)

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